

TEJAS SACHDEVA

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EDUCATION

The University of Texas at Austin – Austin, TX

Master of Science in Mechanical Engineering

May 2027

Bachelor of Science in Mechanical Engineering

May 2026

EXPERIENCE

Biotex, Inc. – Houston, TX

May 2026 – Aug 2026

Mechanical Engineering Intern

- Engineered 5 CNC-machined 6061-T6 aluminum assembly fixtures for motor-driven treatment pod subassemblies in the Oncomagnetics brain cancer therapy helmet, applying DFA, tolerance stack-up analysis, and ASME Y14.5 GD&T to reduce assembly time 70% (5.0 to 1.5 min/unit) across 400+ units; created manufacturing drawings and inspection criteria for critical features
- Reverse-engineered and redesigned the helmet retention system, developing an ergonomic locking dial and parametric 95A TPU head cradle manufactured via MJF to improve fit, comfort, and prototyping speed
- Designed a Delrin fixture for cell culture testing, applying DFM principles and material selection for sterilization and magnetic compatibility

Texas Robotics – Austin, TX

Dec 2025 – Present

Research Assistant

- Engineered precision test fixtures for an advanced lower-limb exoskeleton, enabling high-fidelity human kinematic data capture to inform clinical validation and iterative design
- Designed and 3D-printed carbon fiber load-bearing components meeting a strict 200 Nm/Rad impedance boundary, reducing structural failure risk under dynamic loading conditions and accelerating prototype-to-test cycle

General Motors – Warren, MI

May 2025 – Aug 2025

Service Release Engineering Intern (Chassis, Body Exterior)

- Delivered \$700K+ in cost savings by identifying part commonality across 2023-2025 Equinox/Terrain platforms, enabling BOM simplification and long-term manufacturing alignment across a high-volume vehicle program
- Leveraged Siemens Teamcenter to compare part geometry and critical-to-function features across multiple vehicle lines, synthesizing findings into actionable design recommendations adopted by cross-functional engineering teams
- Partnered with product and supply chain stakeholders to rationalize duplicate components, directly supporting GM's platform consolidation strategy and reducing service lifecycle complexity

Longhorn Racing Solar – Austin, TX

Aug 2024 – Aug 2025

Powertrain Engineer

- Designed battery cooling manifolds in SolidWorks and ran ANSYS Fluent thermal simulations, achieving 25% improvement in temperature uniformity and 35% reduction in pressure drop under race conditions

Parker Wellbore – Houston, TX

May 2024 – Aug 2024

Technical Services Intern

- Designed a specialized crane basket to safely transport heavy BOP stacks on Parker Wellbore rigs, building 3D models and technical drawings in SolidWorks and Autodesk Mechanical
- Performed FEA (linear static, modal, fatigue) per API, ASME, and AISC standards to determine FOS and WLL; selected materials via Charpy V-notch testing, achieving a 65% reduction in Von-Mises stress fluctuation under impact loading

LEADERSHIP AND COMMUNITY INVOLVEMENT

First-Year Experience (UT Austin) – Austin, TX

May 2024 – Jan 2025

First-Year Interest Group (FIG) Mentor

- Mentored a cohort of 20+ first-year students through academic and social transition, delivering individualized guidance recognized by mentees as key to their successful adjustment to college

ASME Purdue University Chapter – West Lafayette, IN (prior to transferring to UT Austin)

Oct 2022 – May 2023

Treasurer

- Managed \$20K+ in funds and secured external sponsorships to support engineering competitions and initiatives

ADDITIONAL

- Technical /Computer Skills:** SolidWorks, Autodesk Fusion 360, AutoCAD Mechanical, GD&T, DFM, DFA, Siemens Teamcenter, ANSYS (Fluent, Mechanical), MATLAB, Python, LabVIEW, DFSS Green Belt
- Honors:** Richard Douglas and Judith Watson Perkins Endowed Presidential Scholar